Global Settlement Classification from True Color Satellite Imagery

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***Abstract*—Accurate and scalable identification of human settlements from satellite imagery is essential for urban planning, disaster response, and sustainable development monitoring. This study presents a global framework for classifying true-color satellite imagery into urban, rural, and uninhabited categories using physically observed Earth observation data. Imagery is obtained from the Harmonized Landsat–Sentinel (HLS) v2.0 collection, providing 30-meter surface reflectance measurements with consistent global coverage. Ground truth labels are derived from the Global Urban and Rural Settlement (GURS) dataset through spatially aligned raster windowing and resolution-aware class weighting. A globally distributed tile dataset spanning six continents is constructed using reprojection, affine coordinate alignment, and systematic filtering of low-information regions. Unsupervised clustering, gradient boosting, and deep convolutional neural networks are evaluated under conditions of extreme spatial class imbalance, reflecting the predominance of uninhabited land. While convolutional networks achieve higher overall accuracy, gradient boosting demonstrates stronger recall for inhabited classes, highlighting trade-offs between model complexity and reliable settlement detection in imbalanced geospatial data.**

***Index Terms*—remote sensing, global settlement mapping, urban–rural classification, satellite imagery, raster preprocessing, spatial class imbalance, machine learning**

I. Introduction

Satellite imagery has become increasingly abundant, with thousands of artificial satellites continuously capturing vast quantities of high-resolution data of the Earth's surface. This volume of imagery contains many spatial and environmental patterns, but manual analysis is very time consuming and inefficient at scale, making machine learning an essential tool for extracting meaningful results. The objective of this research is to evaluate and compare several learning methods in their ability to classify true-color satellite tiles into urban, rural, and uninhabited settlements.

II. Related Works

The automatic detection and classification of human settlements from satellite imagery has been an active area of research, driven by the increasing availability of high-resolution optical and multispectral data. Early efforts focused on supervised and unsupervised machine learning methods to classify built-up areas and urban patterns. Corbane et al. demonstrated the use of convolutional neural networks (CNNs) for global settlement mapping from Sentinel-2 imagery, providing a scalable approach for identifying human settlements at continental scales [7]. Similarly, Qiu et al. developed fully convolutional networks for large-scale human settlement extent (HSE) mapping, showing that deep architectures can segment inhabited regions accurately from multi-spectral imagery [8]. Helber et al. further explored semantic segmentation techniques for generating settlement layers, highlighting the potential of CNNs to capture spatial heterogeneity in urban and rural areas [9]. Attention-based ResNeXt architectures have also been applied to multi-seasonal Sentinel-2 data, improving settlement classification accuracy by modeling spectral and temporal variability [10].

In parallel, the challenges of class imbalance in remote sensing datasets have been investigated. Fonseca, Douzas, and Bacao proposed k-means clustering combined with synthetic minority oversampling to improve detection of minority classes in land cover classification, directly addressing issues where urban or rural settlements are vastly outnumbered by uninhabited regions [6]. Studies in urban change detection and object-based land cover classification emphasize that misclassified low-density settlements or scattered habitation can bias downstream applications, including infrastructure planning and population estimation [8], [9].

Beyond technical modeling, ethical and operational considerations are increasingly recognized as essential in the remote sensing domain. Paolanti et al. established a framework to quantify the trustworthiness of AI in remote sensing, emphasizing transparency, fairness, and accountability [6]. Oluoch examined the ethical implications of AI in geographic information systems, highlighting potential misuse of geospatial data in ways that affect privacy and vulnerable populations [12]. The National Space Council's advisory report also underscores risks associated with satellite-based monitoring, particularly when imagery could be used for unauthorized surveillance or political repression [11]. These perspectives motivate careful consideration of model design, data sourcing, and interpretation in applications such as disaster response and urban planning.

While prior work has demonstrated effective classification of settlements using CNNs and other machine learning techniques, this study aims to compare the performance of both unsupervised and supervised methods, addressing the

computational and operational challenges inherent in large-scale human settlement mapping from satellite imagery.

III. Data and Preprocessing

A. Dataset Collection

The ground truth data for this study are derived from the Global Urban and Rural Settlement (GURS) dataset [4]. This dataset provides comprehensive coverage of urban and rural settlements from 2000 to 2020, offering high-resolution, tile-based mapping of global settlement patterns. Using GURS 2020 [4] ensures that our analyses leverage the most up-to-date and systematically curated settlement data, benefiting from the extensive effort undertaken by Liu et al. to scan and classify settlements across all tiles worldwide.

To ensure that the models are trained on authentic imagery, we restrict our data to true-color satellite products rather than simulated or synthetically generated layers. We obtain our imagery from the Microsoft Planetary Computer [5], which hosts the Harmonized Landsat and Sentinel (HLS) v2.0 collection providing 30-meter multispectral observations with global coverage [2]. This dataset serves as the basis for all experiments, and the following steps describe the acquisition pipeline:

1) For each continent, fetch 100 random queries. Total size estimate $= 6\,\text{continents} \times 100 \times 35\,\text{MB} = 21\,\text{GB}$
2) For each query, download the corresponding TIFs for the RGB color channels.
3) Combine all three channels into a single raster.
4) Save the raster on disk with the file name corresponding to which continent the data was from.
5) Add optional short circuiting to prevent re-downloading when already cached.

B. Region Selection

Equal coverage of most human settlements in the world is very important for training and evaluation, so six bounding boxes will be selected, each representing a geographical area of human activity for every inhabited continent. Europe and Asia will be considered separate continents since analysis is done on human and not natural geography. Some additional selection optimizations:

- For North America, much of northern Canada and all of Alaska will be cut off due to low population density.
- For Asia, much of the Siberian wilderness and the Eurasian Steppe will be omitted due to low population density.
- For South America, most of Patagonia will be cut off due to sparse population density.
- For Africa, most imagery will be selected from a bounding box that focuses on sub-Sahara due to similar reasons as above.
- Australia's outback will be kept to see how the model performs.

Edge cases, such as ocean data and Antarctica, may appear sporadically but are not the main focus of the model. Fig. 1 indicates the coverage of the world by the training dataset.

C. Raster Windowing

The raw GeoTIFF imagery from HLS v2.0 comes in various coordinate reference systems, whereas GURS operates in Mollweide [4]. Accordingly, both GURS and all raw imagery are reprojected to EPSG:4326 to ensure spatial consistency. To facilitate efficient processing and avoid loading entire GeoTIFFs into memory, each raster is subdivided into 64 tiles, which enhances surface-level detail for the model. Tiles that contain predominantly empty regions—defined as having more than 20% black pixels—are discarded. All remaining tiles are saved to a dedicated output directory for subsequent use.

The ground truths for each tile are extracted from the GURS 2020 raster using two principal steps:

1) Isolate the portion of GURS 2020 corresponding to the geographical extent of each tile.
2) Assign a score to each land-cover label in GURS (urban, rural, uninhabited) reflecting the fractional area of that class within the tile.

Let the bounds of a training tile be defined by four latitude-longitude pairs representing the corners of the bounding box, denoted as UL, UR, LL, and LR. The objective is to determine the row and column indices in GURS corresponding to these bounds: $x \in [x_0, x_f]$, $y \in [y_0, y_f]$.

Only two degrees of freedom (UL and LR) are required to define the full bounds. The affine transformation of the raster is used to map the tile coordinates to row and column indices:

$$\mathbf{T} = \begin{bmatrix} 0 & 0 & -180 \\ 0 & 0 & 77.91 \\ 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

Apply the inverse to get row and column bounds:

$$\mathbf{T}^{-1}\,UL = \begin{bmatrix} x_0 \\ y_0 \\ 1 \end{bmatrix}, \quad \mathbf{T}^{-1}\,LR = \begin{bmatrix} x_f \\ y_f \\ 1 \end{bmatrix}, \tag{2}$$

where y represents the rows and x represents the columns. An additional processing step is needed to ensure $x, y \in \mathbb{Z}$:

$$\begin{aligned} x_0 &= \lfloor \min(x_0, x_f) \rfloor, \quad x_f = \lfloor \max(x_0, x_f) \rfloor, \\ y_0 &= \lfloor \min(y_0, y_f) \rfloor, \quad y_f = \lfloor \max(y_0, y_f) \rfloor. \end{aligned} \tag{3}$$

Let $GURS_{wnd} = GURS[y_0 : y_f, x_0 : x_f]$ represent the slice of the GURS raster at the defined bounds. After following these steps above, we get $GURS_{wnd} \in \{1, 2, 127\}^{126 \times 148}$. This is slightly smaller than the training tile's dimensions because of difference in resolution.

D. Computing Relative Class Frequencies

Define a binary mask for a value n:

$$\mathcal{M}^{(n)}(i,j) = \begin{cases} 1, & GURS_{wnd}[i][j] = n, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

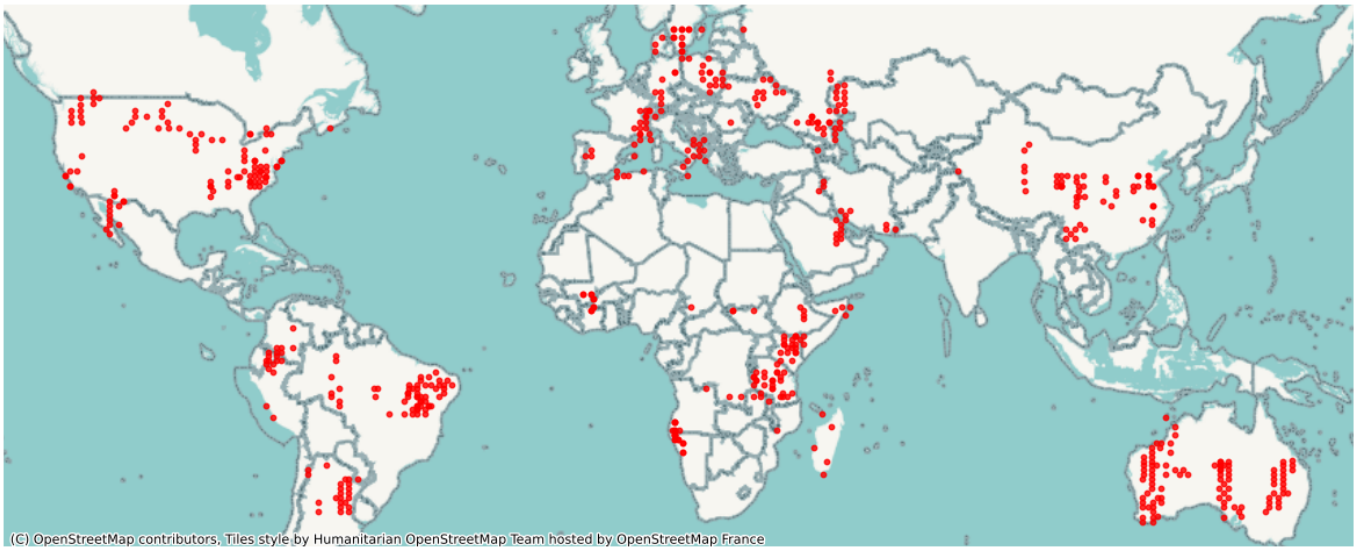


Fig. 1. Global map showing the locations of selected satellite tiles, with each red dot indicating a region from which imagery is sampled.

To get the frequency of any value n, define

$$\mathcal{F}(n) = \frac{\sum_{i=y_0}^{y_f} \sum_{j=x_0}^{x_f} \mathcal{M}^{(n)}(i,j)}{126 \times 148 = 18648}. \tag{5}$$

Next, the weights of each inhabited class are set to $100/9$. This is because each GURS pixel is $100 \times 100 = 10000$ m^2 whereas the tile pixels are only $30 \times 30 = 900$ m^2. The ratio of the weights represents the size differences of the pixel. Uninhabited pixels do not get weighted as the population density is negligible. We use the following function to ensure normalized weighted frequencies:

$$F_w(n) = \frac{w_n \mathcal{F}(n)}{\sum_k w_k \mathcal{F}(k)}. \tag{6}$$

IV. Methodology

We outline the training procedures and tuning strategies applied to each of our candidate models. Although all models are trained using soft probability distributions as their target outputs, it is assumed that evaluation proceeds using discrete class assignments. Specifically, a predicted label is obtained by computing argmax over the model's soft probability vector and converting the result into a one-hot encoded representation by setting the selected class entry to 1 and all others to 0.

A. Unsupervised Clustering

For k-means clustering, we reduce each image tile $\mathbf{I}_i \in \mathbb{R}^{3\times H\times W}$ to a compact numerical representation. This process should allow k-means to easily cluster the data based on certain visual aspects of the tile. For example, urban tiles are more noisy due to the number of structures. Uninhabited deserts should look smooth and homogeneous.

For each RGB band, mean, standard deviation, $10^{\text{th}}, 50^{\text{th}}$, and 90th percentile statistics were computed. To capture intra-tile heterogeneity, we convert the tile to grayscale:

$$\begin{aligned}\mathbf{I}_{i,\text{gray}} = 0.2989\,\mathbf{I}_i[\text{red}] + 0.587\,\mathbf{I}_i[\text{green}] \\ + 0.114\,\mathbf{I}_i[\text{blue}].\end{aligned} \tag{7}$$

These coefficients are derived from the ITU-R BT.601 (Rec. 601) standard [1], which defines how to convert RGB (in the sRGB color space) to luma, i.e., perceived brightness. We then can compute a noise statistic based on the gradient magnitude using the Sobel operator:

$$\sigma_{i,\text{noise}} = \sqrt{\mathbb{E}\Big[\big(|\nabla \mathbf{I}_{i,\text{gray}}| - \mu_{i,\text{noise}}\big)^2\Big]}. \tag{8}$$

Lastly, we can capture local heterogeneity by dividing the grayscale image into $n \times n$ patches $\{P_k\}_{k=1}^{n^2}$, and computing the variance within each patch:

$$\sigma^2_{i,\text{patch}} = \frac{1}{n^2} \sum_{k=1}^{n^2} (\text{Var}(P_k) - \mu_{i,\text{patch}})^2. \tag{9}$$

The final feature vector $\vec{x_i} \in \mathbb{R}^{1\times 19}$ combines the percentiles, as well as the means and standard deviations of both the noise and patch statistics. Optimal k was selected via silhouette score as shown in Fig. 7.

B. Feature-Based Learning Model

For gradient boosting, we once again reduce each image tile $\mathbf{I}_i \in \mathbb{R}^{3\times H\times W}$ to a compact numerical representation. For each band $b \in \{\text{red, green, blue}\}$, define the following statistics over all pixels x, y in b:

$$\mu_{i,b} = \frac{1}{HW} \sum_{y=0}^{H-1} \sum_{x=0}^{W-1} \mathbf{I}_i[b][x][y], \tag{10}$$

TABLE I
OPTIMAL GRADIENT BOOSTING HYPERPARAMETERS AFTER GRID SEARCH. LOG LOSS ≈ 0.142.

Hyperparameter	Value
`colsample_bytree`	0.8
`n_estimators`	800
max depth	8
η	0.01
α	0
λ	1
subsample	0.8

$$\sigma_{i,b} = \sqrt{\frac{1}{HW}\sum_{y=0}^{H-1}\sum_{x=0}^{W-1}(\mathbf{I}_i[b][x][y] - \mu_{i,b})^2}, \quad (11)$$

$$\min(\mathbf{I}_i[b]), \quad \max(\mathbf{I}_i[b]), \quad (12)$$

where $0 \leq i \leq N$ and H is the height, W is the width in pixels, and $N = 30444$ is the cleaned dataset size. This totals to 12 features.

The target matrix $\mathbf{y} \in \mathbb{R}^{N\times 3}$ has each row $\vec{y_i} = \begin{bmatrix} p_{\text{urban}} & p_{\text{rural}} & p_{\text{uninhabited}} \end{bmatrix}$ representing a set of probabilities of each class being the dominant settlement type for each tile. As mentioned earlier, we derive discrete one-hot class labels via $t_i = \arg\max_c \vec{y_i}[c]$, where $t_i \in \{\text{urban, rural, uninhabited}\}$. Due to extreme class imbalance, we also want to oversample minority classes to give more representation and balance out the dataset. Hyperparameters were selected using K-fold cross-validation over a grid search. Optimal hyperparameters are displayed in Table I.

C. Deep Learning Architecture

For deep learning evaluation, convolutional neural networks were selected due to their robust image processing capabilities. Specifically, ResNet-34 [3] was used due to hardware constraints of the testbed. To convert a raw tile into the format that ResNet-34 expects, we employ the following steps:

1) Load the raw tile from disk into $\mathbf{X}_i^{\text{raw}} \in \mathbb{R}^{3\times H\times W}$ where $H \times W$ are the original spatial dimensions of the tile.
2) Normalize to $[0, 1]$:

$$\mathbf{x}_i^{\text{norm}} = \frac{\mathbf{X}_i^{\text{raw}}}{255} \in [0,1]^{3\times H\times W}. \quad (13)$$

3) Finally, resize to a fixed input shape using bilinear interpolation:

$$\mathbf{x}_i = \text{resize}(\mathbf{x}_i^{\text{norm}}) \in \mathbb{R}^{3\times 256\times 256}. \quad (14)$$

Similar to gradient boosting, we employ a $K = 5$ fold cross-validation selection scheme. Oversampling the minority datasets was also done to account for class imbalance. The best hyperparameters found after a 5-fold cross validation on a grid of 18 candidates are shown in Table II.

TABLE II
OPTIMAL RESNET-34 HYPERPARAMETERS AFTER GRID SEARCH. CROSS-ENTROPY LOSS ≈ 0.3195.

Hyperparameter	Value
η	0.0001
λ	10^{-5}
p	0.2

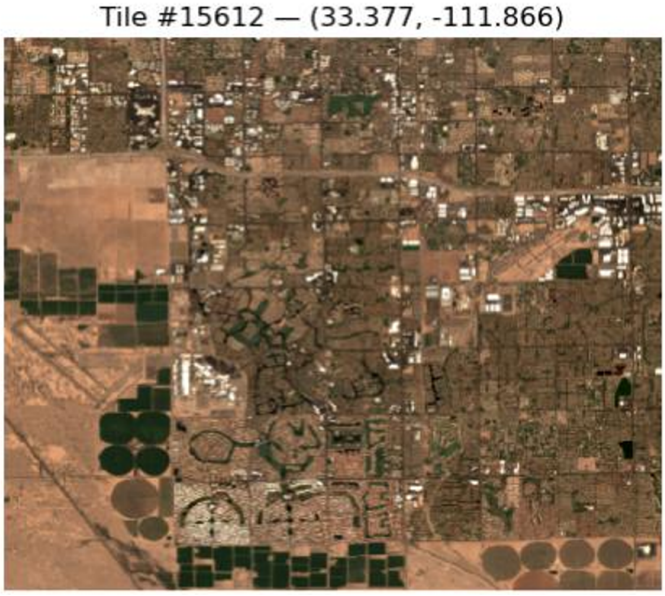


Fig. 2. True-color satellite photograph of Chandler, U.S. (part of greater Phoenix metropolitan area).

V. RESULTS

ResNet-34 outperformed gradient boosting on all selected metrics except for loss and macro recall. For all metrics shown in Table III except loss, higher values indicate better performance. While these metrics could likely be improved with additional training, model tuning, or larger datasets, the inherent characteristics of satellite tiles impose a natural performance barrier that may limit further gains.

Fig. 2 shows the tile representing the highest-accuracy prediction by gradient boosting. The ground truth value is $p_{\text{urban}} \approx 0.98$ whereas gradient boosting predicted $p_{\text{urban}} \approx 0.96$. The top urban tile prediction for ResNet-34 is shown in Fig. 3 (ground truth $p_{\text{urban}} \approx 0.95$, predicted $p_{\text{urban}} \approx 0.96$).

Although both gradient boosting and ResNet-34 had similar performance on urban tiles, they diverge in performance for rural tiles. Fig 4 represents the top prediction for rural tiles made by gradient boosting. Even though the ground truth $p_{\text{rural}} \approx 0.62$, gradient boosting overestimated p_{rural} to be 0.88. ResNet-34's prediction for Fig. 5 underestimated p_{rural} to be 0.66, even though the ground truth for the tile is $p_{\text{rural}} \approx 0.72$.

A. Validation

For k-means learning, silhouette score was chosen as the validation metric due to the extremely insignificant within-cluster sum of squares: $\log_{10}(\text{WCSS}) \approx -9$. This makes it



Fig. 3. True color satellite photograph of western Berlin, Germany.



Fig. 4. True color satellite photograph of Linying County, China.

very difficult to determine the optimal number of clusters using the elbow method. Switching to using silhouette scores for validation makes the optimal k much more apparent (Fig. 7).

The decision for validation was reporting macro-averaged recall, precision, and F1 score rather than the standard (micro) metrics. Macro-averaging treats each class equally, regardless of its prevalence, which is particularly important here because uninhabited tiles vastly outnumber inhabited ones as mentioned previously. A model with high recall ensures that most true settlements are detected rather than overlooked. In contrast, if we were to focus on overall accuracy or precision, we could fail to detect certain types of settlements, such as small, remote, or marginalized communities. Prioritizing recall therefore acts as a safeguard against creating these blind spots that often mask underrepresented populations.

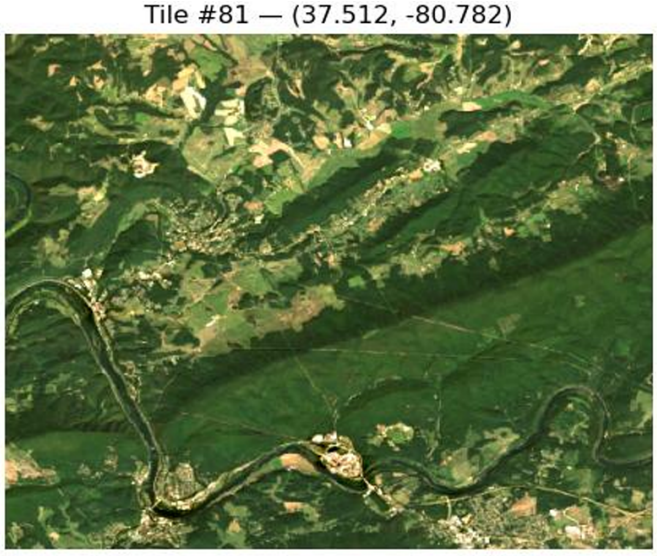


Fig. 5. True color satellite photograph of the southeastern border of West Virginia, U.S.

TABLE III
COMPARISON OF GRADIENT BOOSTING AND RESNET-34 ACROSS KEY VALIDATION METRICS.

Metric	XGBoost	ResNet-34
Loss	0.2932	0.3172
Accuracy	0.8796	0.9629
Macro Precision	0.4305	0.6965
Macro Recall	0.6284	0.5074
Macro F1 Score	0.4686	0.5653

VI. ANALYSIS

For the unsupervised model, we can see that from Fig. 6, despite having statistics that quantify noise, k-means still primarily clustered on color. Cluster 0 mainly has tiles of warmer hues, such as deserts, red soil, and dry vegetation. Cluster 1 is composed mainly of green forestry and grassland tiles, which may indicate a higher presence of human settlements. Cluster 2 corresponds to water tiles, characterized by a distinct and uniform blue texture. Rather than clustering based on human settlements, k-means clustering partitioned the tiles according to biome. Based on Fig. 7, the silhouette score tells us that the clusters are reasonably separated, but there is some overlap, which is not excellent, but typical for real-world, noisy, high-dimensional data like satellite tiles. Fig. 6 displays some sample tiles from each cluster to visualize the distinction between each cluster.

This outcome highlights certain limitations of k-means when applied to settlement detection. Biomes with similar spectral characteristics can be easily confused; for example, dry grasslands and sparsely vegetated urban borders exhibit comparable reflectance, leading the algorithm to group them together despite differences in land use. Moreover, the distinction between rural and urban regions is often ambiguous. Rural areas frequently contain scattered settlements amidst vegeta-

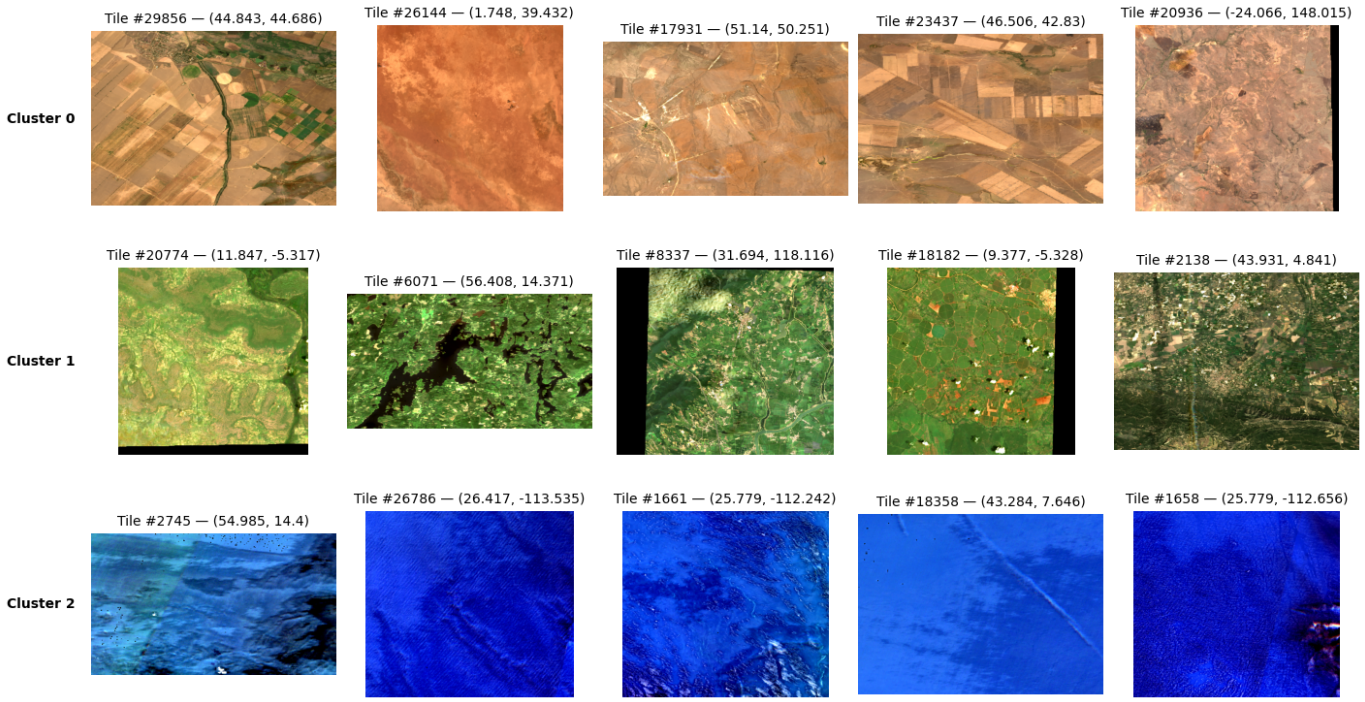


Fig. 6. Sample visualization of the three clusters, shown over representative satellite tiles arranged in a grid. Each row illustrates characteristic imagery associated with one of the identified clusters.

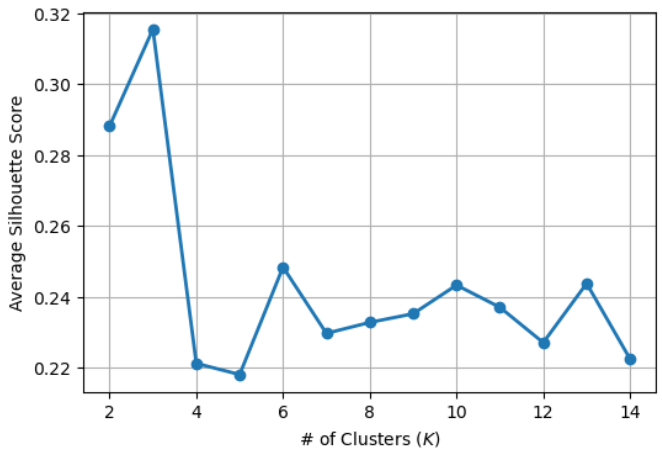


Fig. 7. Mean Silhouette Score vs. Number of Clusters

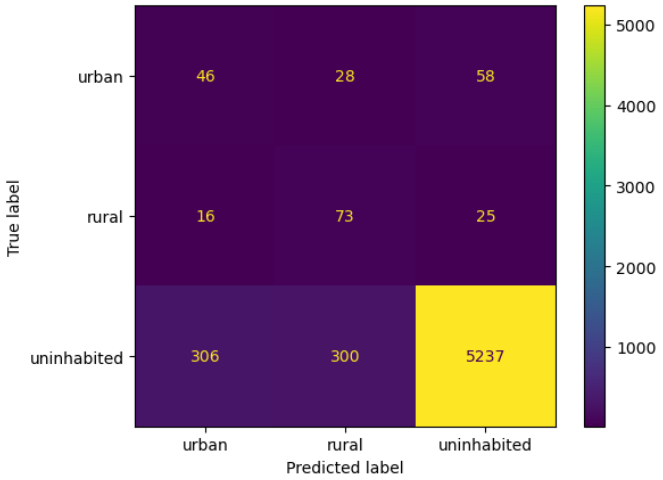


Fig. 8. Confusion matrix for gradient boosting predictions on the test set.

tion, while suburban or urban zones display mixed signatures of vegetation, soil, and built structures. Tiles spanning these transitional areas are particularly prone to misclassification, resulting in false negatives where human presence exists but is not captured by the dominant color-based clustering. These results may open opportunities for applications in environmental monitoring, ecological conservation, biodiversity assessment, land-use planning, climate impact studies, and natural resource management.

For both supervised learning models, the confusion matrices makes it abundantly clear that uninhabited tiles outweigh urban and rural tiles even minority oversampling is done. From both Fig. 8 and Fig. 9, the true positives for uninhabited tiles are so numerous that the other predictions are negligible by comparison. Such an imbalance is inevitable and present in virtually all satellite imagery datasets, as the proportion of land covered by human settlements is minuscule compared to uninhabited areas. Importantly, this imbalance is extreme even after sampling and excluding obviously uninhabited regions through bounding box selection.

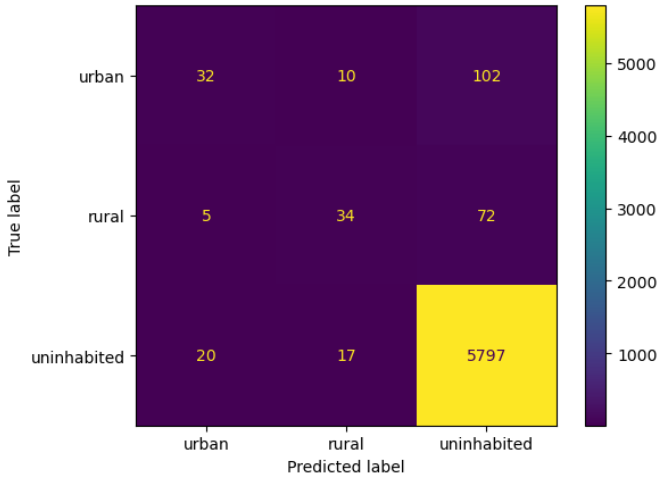


Fig. 9. Confusion matrix for ResNet-34 predictions on the test set.

VII. Ethical Discussion

Developing models that identify human settlements carries an inherent risk of misuse, even when the intended purpose is to support beneficial domains such as urban planning, emergency rescue, and disaster response. The same digital system responsible for designing better land-use could also be repurposed for harmful goals, such as unauthorized population tracking, nation-state surveillance, political repression, or selective resource denial. Because the underlying algorithms scale easily once developed, the ethical responsibility lies in recognizing that precise insights from satellite imagery can generate both constructive and destructive actions depending on who controls them.

A second ethical consideration involves responsible and transparent data sourcing, particularly when using large-scale satellite imagery datasets. While these datasets are typically open and designed for research, the obligation remains to ensure that the imagery complies with licensing terms, respects privacy expectations for publicly captured scenes, and does not inadvertently include sensitive or restricted content. Ethical sourcing also requires acknowledging the limitations of satellite imagery, such as resolution constraints, atmospheric artifacts, or temporal gaps, to avoid overstating what the model can reliably infer. Being transparent ensures that the models are explainable, understandable, and interpretable to non-technical stakeholders [6].

Finally, even as these models become highly performant and precise, the final screening should always be done by a human operator. When dealing with human settlements, a single mistake could impact not only individual lives but entire communities and cultures. Not all human settlements are built equally, and we must be careful in generalizing thousands of unique lifestyles, traditions, cultures, and heritage when applying a machine learning model to predict a class label.

VIII. Conclusion

Based on our experimental results, we recommend *gradient boosting* as the preferred supervised model for this satellite tile dataset. Gradient boosting demonstrates several advantages: it is computationally lighter than deep learning alternatives, making it more feasible for large-scale deployment, and it achieves lower loss and higher recall, which is particularly important given the extreme class imbalance in our data.

Potential improvements for future work include integrating temporal satellite data to capture dynamic changes, incorporating higher-resolution imagery for detecting small or informal settlements, and combining gradient boosting with unsupervised embeddings to better generalize across different geographic regions. Such extensions could expand the model's applicability to scenarios that have not yet been fully explored, including predicting settlement growth, monitoring environmental impacts of urbanization, or supporting policy decisions in undermapped regions.

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